THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonals of a rhombus are 18 cm and 80 cm. Its area is: 1

(a) 720 cm^2 **(b)** 800 cm^2 **(c)** 730 cm^2 **(d)** 738 cm^2

2. The diagonal of a rectangle is 29 cm and one side is 20 cm. Its area is: 1

(a) 420 cm^2 **(b)** 441 cm^2 **(c)** 440 cm^2 **(d)** 430 cm^2

3. In a parallelogram, two opposite angles are $(2x + 5)^\circ$ and $(x + 71)^\circ$. The value of x is: 1

(a) 68 **(b)** 66 **(c)** 67 **(d)** 65

4. The diagonals of a quadrilateral are 87 cm and 120 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 416 cm **(b)** 207 cm **(c)** 621 cm **(d)** 310 cm

5. The four angles of a quadrilateral are in AP with common difference 22°. The smallest angle is: 1

(a) 67° **(b)** 79° **(c)** 35° **(d)** 57°

6. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 20 cm. Its perimeter is: 1

(a) 60 cm **(b)** 56 cm **(c)** 84 cm **(d)** 76 cm

7. The diagonal of a square is $12\sqrt{2}$ cm. Its perimeter is: 1

(a) 56 cm **(b)** 60 cm **(c)** 72 cm **(d)** 48 cm

8. A rhombus has side 41 cm and one diagonal 18 cm. The other diagonal is: 1

(a) 80 cm **(b)** 120 cm **(c)** 78 cm **(d)** 82 cm

9. **Assertion (A):** In $\triangle ABC$, D and E are the midpoints of AB and AC. If $BC = 10$ cm, then $DE = 5$ cm. 1

Reason (R): The line segment joining the midpoints of two sides of a triangle is parallel to the third side and half of it.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** In a parallelogram, if one angle is 70°, then each adjacent angle is 110°. 1

Reason (R): The adjacent angles of a parallelogram are supplementary.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In parallelogram ABCD, $AB = (2x + 8)$ cm, $CD = (x + 35)$ cm and $BC = (x + 1)$ cm. Find x, the sides and the perimeter. 2

12. A rhombus has side 41 cm and one diagonal 18 cm. Find the other diagonal, its area and the distance between a pair of opposite sides (height), correct to two decimal places. 2

13. The angles of a quadrilateral are in AP with common difference 22°. Find all four angles and say whether a pair of them is supplementary. 2

14. In parallelogram ABCD, $AB = (3x + 3)$ cm, $CD = (x + 67)$ cm and $BC = (x + 6)$ cm. Find x, the sides and the perimeter. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The diagonals of a parallelogram ABCD meet at O. $OA = (4x + 8)$ cm and $OC = (3x + 36)$ cm. Find x and the length of AC. 3

16. In rhombus ABCD, $\angle A = 110^\circ$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

17. In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 14$ cm and $AB = 18$ cm. Find DE, AD and show that E is the midpoint of AC. 3

OR

The diagonals of a parallelogram ABCD meet at O. $OA = (5x + 1)$ cm and $OC = (3x + 35)$ cm. Find x and the length of AC. 3

18. In rhombus ABCD, $\angle A = 128^\circ$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

OR

In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 26$ cm and $AB = 26$ cm. Find DE, AD and show that E is the midpoint of AC. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Case study. A triangular park ABC has $AB = 18$ m, $BC = 24$ m and $CA = 26$ m. A pole D is at the midpoint of AB and a pole E is at the midpoint of AC. A rope is stretched from D to E. 5

(i) Find the length of the rope DE.

(ii) How is DE related to BC in direction?

(iii) Find AD and AE.

(iv) Find the perimeter of $\triangle ADE$ and compare it with the perimeter of $\triangle ABC$.

20. The midpoints of the sides of a rectangle 40 cm by 42 cm are joined in order. Show that the figure formed is a rhombus and find its perimeter and area. 5

OR

In a parallelogram ABCD the diagonals meet at O with $OA = 9$ cm, $OB = 12$ cm and $AB = 15$ cm. Show that ABCD is a rhombus and find its perimeter. 5

END OF QUESTION PAPER — Total: 40 Marks